

ECON 609 - Problem Set 2

Fall 2022

Be sure to include all the material that shows your work and to present your results in a way that is easy to understand.

Projection and Perturbation

The remaining questions will be based on the following model. There is a representative household with preferences:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} 0.96^t \left(\log c_t - \frac{l_t^2}{2} \right)$$

The household consumes, saves, and works, with a budget constraint

$$c_t + i_t = w_t l_t + r_t k_t$$

There is a production function

$$c_t + i_t = e^{z_t} k_t^{\alpha_t} l_t^{1-\alpha_t}$$

with a law of motion for capital

$$k_{t+1} = 0.9k_t + i_t$$

where $i_t \geq 0$ and a technology level z_t that follows a Markov Chain that takes values in

$$z_t \in \{-0.0673, -0.033, 0, 0.0336, 0.0673\}$$

with transition matrix

$$\begin{bmatrix} 0.9727 & 0.0273 & 0 & 0 & 0 \\ 0.0041 & 0.9806 & 0.0153 & 0 & 0 \\ 0 & 0.0082 & 0.9836 & 0.0082 & 0 \\ 0 & 0 & 0.0153 & 0.9806 & 0.0041 \\ 0 & 0 & 0 & 0.0273 & 0.9727 \end{bmatrix}$$

and $\alpha_t \in \{0.25, 0.3, 0.35\}$ with transition matrix

$$\begin{bmatrix} 0.9 & 0.07 & 0.03 \\ 0.05 & 0.9 & 0.05 \\ 0.03 & 0.07 & 0.9 \end{bmatrix}$$

You can think about α_t as the reduced form of some technological change that modifies the elasticity of output with respect to capital.

(1) Projection

Compute the solution to the previous model when you use 8 Chebyshev polynomials on capital. Plot the policy functions at $z = 0$ and $\alpha = 0.3$. Plot the Euler equation errors at $z = 0$ and $\alpha = 0.3$.

(2) Perturbation

Compute the solution to the previous model using a third-order approximation and plot the IRFs initialized at the mean of the ergodic distribution. You need to substitute the process for productivity by

$$z_t = 0.95z_{t-1} + 0.005\epsilon_t^1$$

where ϵ_t^1 is a normalized Gaussian innovation and the process for α_t by

$$\alpha_t = 0.03 + 0.9\alpha_{t-1} + 0.01\epsilon_t^2$$

where ϵ_t^2 is a normalized Gaussian innovation.